PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

CYBERSECURITY

Pentester Name	Tukadiya Smruti
Post	Cybersecurity beginner
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.
All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the `networkwalks.com` domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.0.4** and **WP Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

I used **Curl** with the `-I` option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint `/wp-json/`.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows `ipconfig` command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified four live hosts:

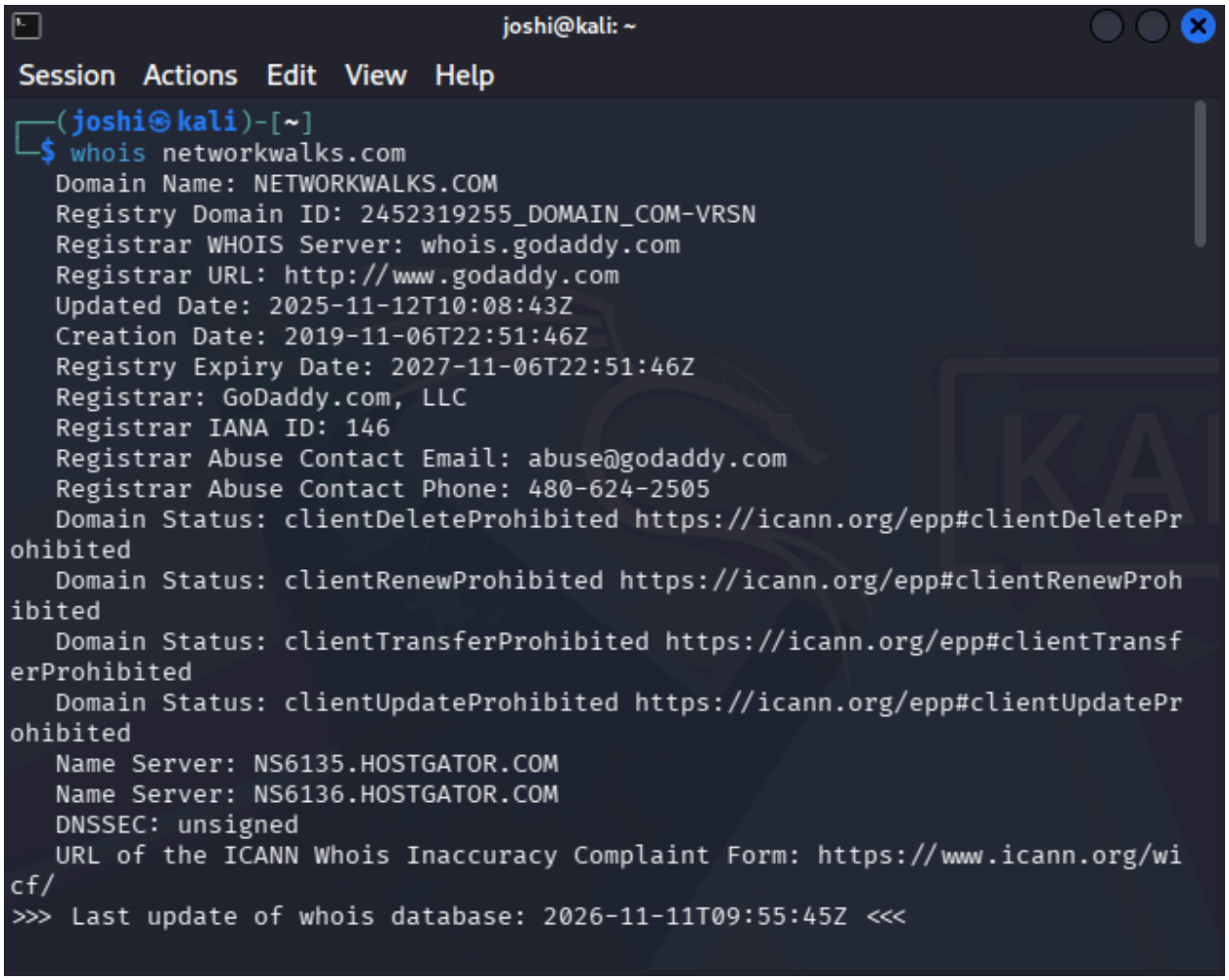
- `10.0.0.1`
- `10.0.0.4`
- `10.0.0.19`
- `10.0.0.5`

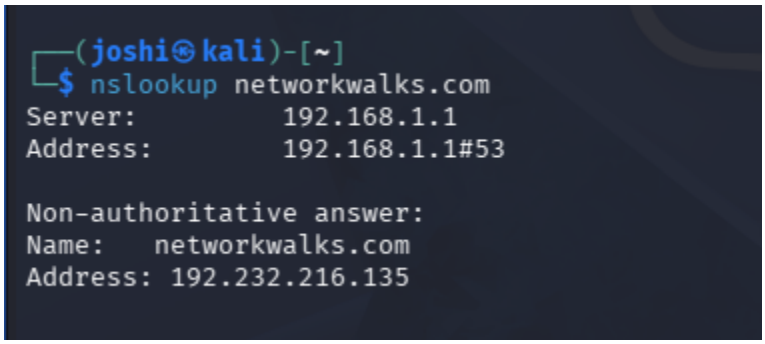
The example results also included four MAC addresses.

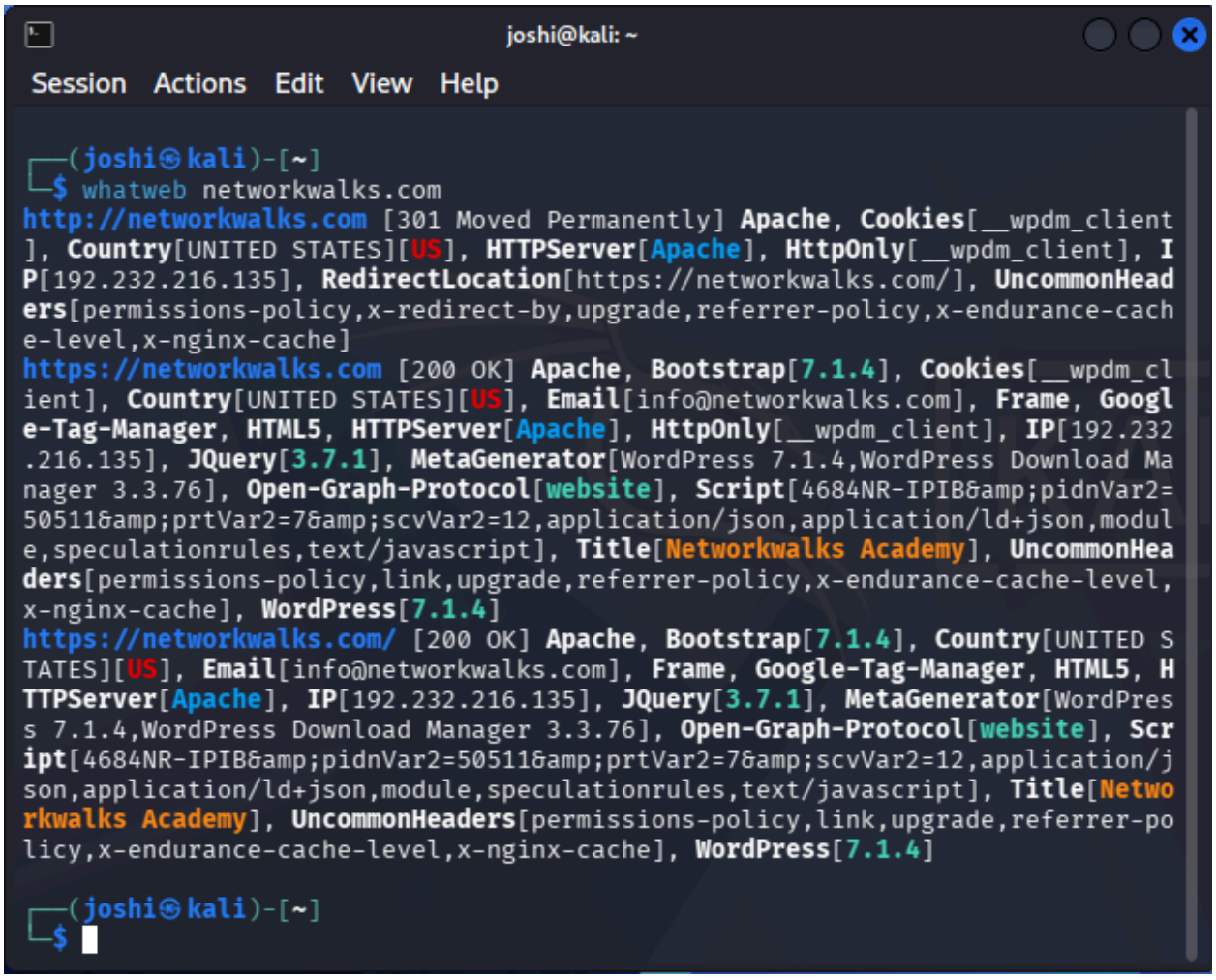
After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

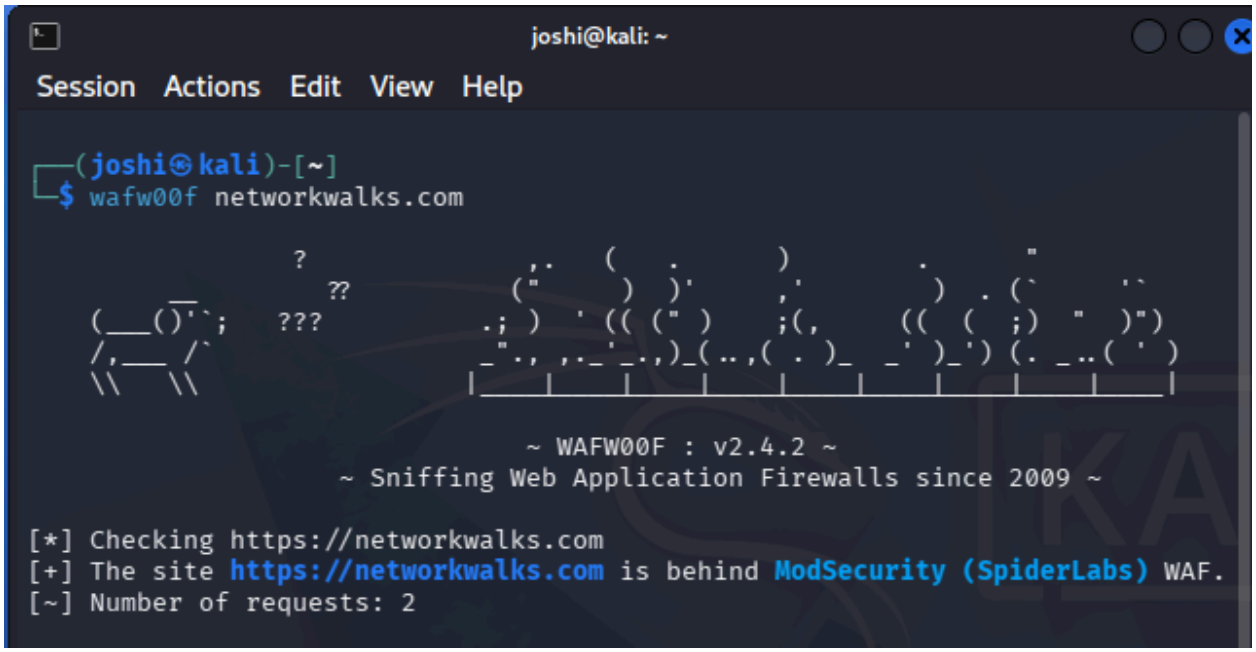
Note: The actual subnet, number of hosts and addresses should be replaced with the results from my own network when submitting the report.

Evidence Collected:

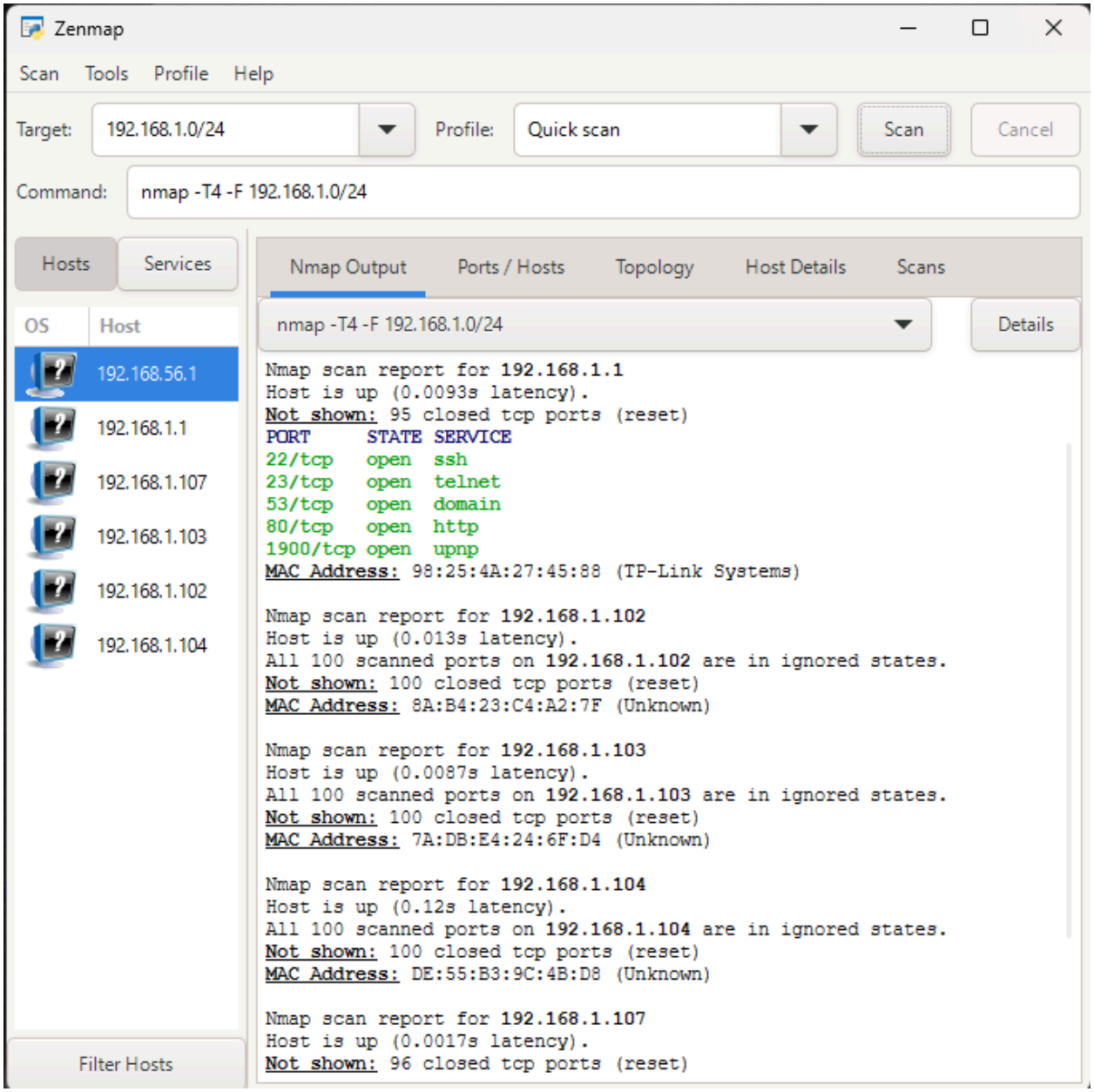


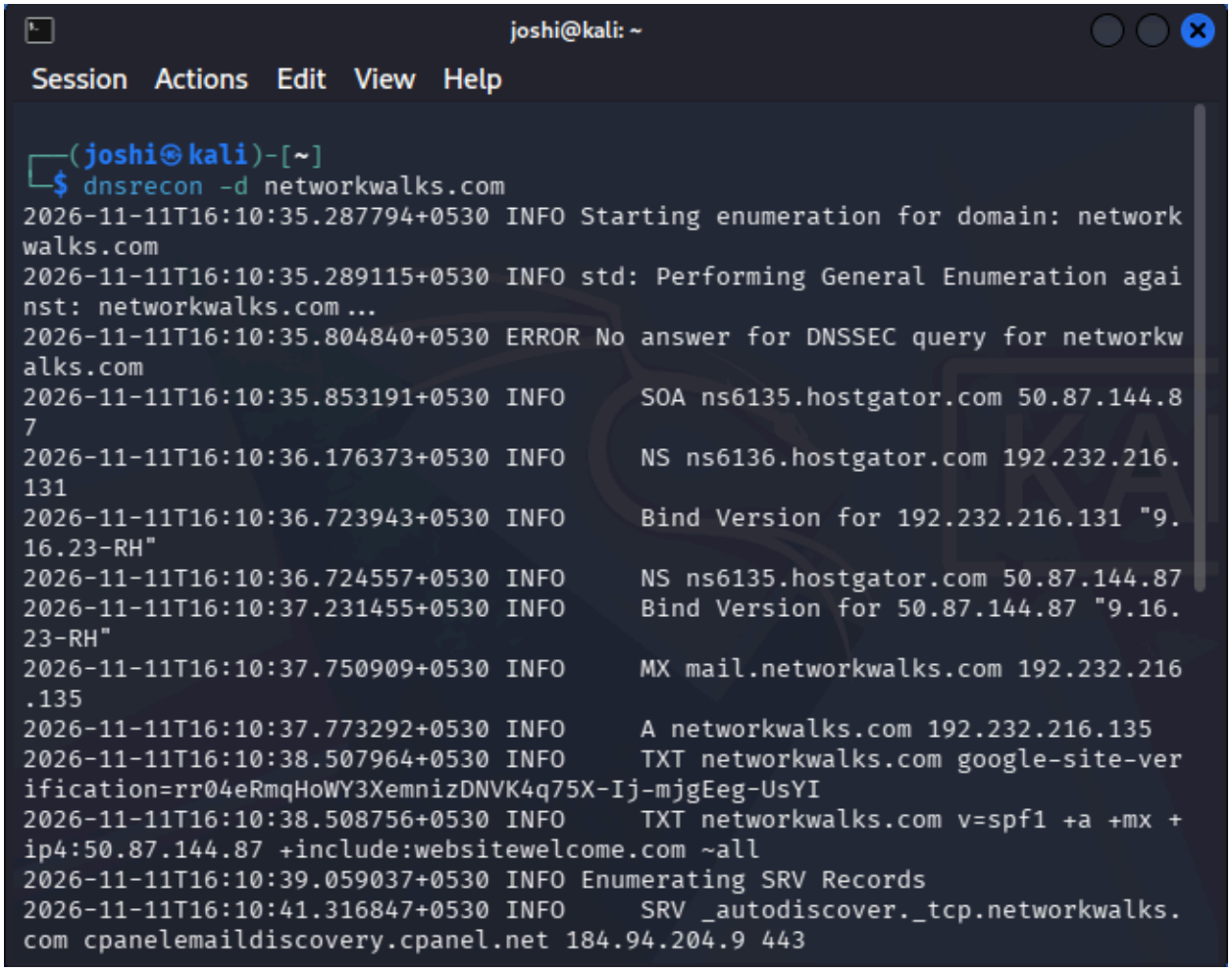



joshi@kali: ~
Session Actions Edit View Help
┌──(joshi㉿kali)-[~]
└─$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
┌──(joshi㉿kali)-[~]
└─$


joshi@kali: ~
Session Actions Edit View Help
┌──(joshi㉿kali)-[~]
└─$ wafw00f networkwalks.com
~ WAFW00F : v2.4.2 ~
~ Sniffing Web Application Firewalls since 2009 ~
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

```
┌──(joshi㉿kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=3e6ea9d2b0eb6bceeb5349de9c91d377; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 11 Nov 2026 10:34:04 GMT
server: Apache
```


Zenmap
Scan Tools Profile Help
Target: 192.168.1.0/24
Profile: Quick scan
Scan
Cancel
Command: nmap -T4 -F 192.168.1.0/24
Hosts
Services
OS
Host
192.168.56.1
192.168.1.1
192.168.1.107
192.168.1.103
192.168.1.102
192.168.1.104
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -T4 -F 192.168.1.0/24
Details
Nmap scan report for 192.168.1.1
Host is up (0.0093s latency).
Not shown: 95 closed tcp ports (reset)
PORT STATE SERVICE
22/tcp open ssh
23/tcp open telnet
53/tcp open domain
80/tcp open http
1900/tcp open upnp
MAC Address: 98:25:4A:27:45:88 (TP-Link Systems)
Nmap scan report for 192.168.1.102
Host is up (0.013s latency).
All 100 scanned ports on 192.168.1.102 are in ignored states.
Not shown: 100 closed tcp ports (reset)
MAC Address: 8A:B4:23:C4:A2:7F (Unknown)
Nmap scan report for 192.168.1.103
Host is up (0.0087s latency).
All 100 scanned ports on 192.168.1.103 are in ignored states.
Not shown: 100 closed tcp ports (reset)
MAC Address: 7A:DB:E4:24:6F:D4 (Unknown)
Nmap scan report for 192.168.1.104
Host is up (0.12s latency).
All 100 scanned ports on 192.168.1.104 are in ignored states.
Not shown: 100 closed tcp ports (reset)
MAC Address: DE:55:B3:9C:4B:D8 (Unknown)
Nmap scan report for 192.168.1.107
Host is up (0.0017s latency).
Not shown: 96 closed tcp ports (reset)



-End-

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📌 **Project Information**
Network Scanning & Information Gathering Using Zenmap and Kali linux
| **Repository:** GitHub